



OLLSCOIL NA hÉIREANN MÁ NUAD
THE NATIONAL UNIVERSITY OF IRELAND MAYNOOTH

MATHEMATICAL PHYSICS

Year 3

Semester 1

2015 - 2016

Quantum Mechanics I
MP363

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Time allowed: $1\frac{1}{2}$ hours

Answer **two** questions

All questions carry equal marks

1. In a classical hydrogen atom, an electron (mass m_e and charge $-e$) moves around a proton (mass m_p and charge $+e$) in a circular orbit of radius r . The laws of motion thus require that the following force-balancing equation must be satisfied:

$$\frac{\mu v^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

where v is the electron's speed, $\mu = m_e m_p / (m_e + m_p)$ is the system's reduced mass and ϵ_0 is the permittivity of free space. Bohr's hypothesis was that the electron's orbital angular momentum, $L = \mu v r$, is quantised in integer multiples of the reduced Planck constant, i.e. $L_n = n\hbar$ for $n = 1, 2, \dots$

- (a) The electron's total energy is

$$E = \frac{1}{2}\mu v^2 - \frac{e^2}{4\pi\epsilon_0 r}.$$

Use this and the above information to show that the quantised energy levels for the atom are

$$E_n = -\frac{\mu e^4}{32\pi^2 \epsilon_0^2 \hbar^2} \frac{1}{n^2}.$$

[10 marks]

- (b) In this Bohr atom, the orbital radius and speed of the electron are also quantised; find expressions for r_n and v_n . [20 marks]
- (c) Explain why μ may be closely approximated by m_e in all of the above expressions and thus find numerical values for E_1 (in electron-volts), r_1 (in angströms) and v_1 (as a percentage of the speed of light). [20 marks]

2. Let $\mathcal{H}$ be a Hilbert space with inner product $\langle | \rangle$.

- (a) Suppose we have a set of n vectors $\{|\phi_1\rangle, \dots, |\phi_n\rangle\}$ in $\mathcal{H}$; explain what is meant if we say this is an *orthonormal* set of vectors. [5 marks]
- (b) Suppose $|\psi\rangle \in \mathcal{H}$ is a linear combination of the orthonormal vectors given in (a), namely,

$$|\psi\rangle = \sum_{i=1}^n c_i |\phi_i\rangle.$$

Show that the coefficients are given by $c_i = \langle \phi_i | \psi \rangle$. [9 marks]

- (c) Let $\mathcal{H}$ be the space of functions defined on the interval $[-L, L]$ with the inner product

$$\langle \psi | \zeta \rangle = \int_{-L}^L \psi^*(x) \zeta(x) dx.$$

Show that the three functions

$$\begin{aligned} \phi_0(x) &= \frac{1}{\sqrt{2L}}, \\ \phi_1(x) &= \frac{1}{\sqrt{L}} \sin\left(\frac{\pi x}{L}\right), \\ \phi_2(x) &= \frac{1}{\sqrt{L}} \cos\left(\frac{\pi x}{L}\right), \end{aligned}$$

form an orthonormal set.

[36 marks]

3. Consider the one-dimensional quantum-mechanical simple harmonic oscillator, where $\{|n\rangle | n = 0, 1, \dots\}$ are the orthonormalised eigenstates whose energies are $E_n = \hbar\omega(n + 1/2)$. The creation and annihilation operators are defined respectively as

$$a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(X - \frac{iP}{m\omega} \right), \quad a = \sqrt{\frac{m\omega}{2\hbar}} \left(X + \frac{iP}{m\omega} \right),$$

where X and P are the position and momentum operators. These act on the eigenstates via

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle, \quad a |n\rangle = \sqrt{n} |n-1\rangle.$$

Suppose this oscillator is in the state $|\psi\rangle$ given by

$$|\psi\rangle = \frac{i}{2} |0\rangle - \frac{\sqrt{3}}{2} |1\rangle.$$

- If the energy of this state is measured, what are the possible results, and with what probabilities do they occur? **[20 marks]**
- Find the expectation value of the Hamiltonian H . **[10 marks]**
- Compute the expectation values of the position operator X and the momentum operator P . (**Hint:** write these operators in terms of $a^\dagger$ and a .) **[20 marks]**

Some Useful Numbers

- Fundamental unit of charge $e = 1.60 \times 10^{-19}$ C
- Electron mass $m_e = 9.11 \times 10^{-31}$ kg
- Reduced Planck constant $\hbar = 1.05 \times 10^{-34}$ J · s
- Permittivity of free space $\epsilon_0 = 8.85 \times 10^{-12}$ C² · N⁻¹ · m⁻²
- Speed of light $c = 3.00 \times 10^8$ m · s⁻¹
- 1 eV = 1.60×10^{-19} J
- 1 Å = 10^{-10} m